

Stationary-polarized trace defect, replica closure, and the correlated

TECT verification-first repository
claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION
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1. Purpose and scope: round one prior-cycle correction

R-123 reduced the direct R-093 source-action problem to the signed trace excess

$$S_h = D_{0,h} - \|b_h\|^2 = \mathbb{E}_0\{\Theta_h - \|\Phi_h\|^2\}, \quad (1.1)$$

after all six rows, contraction-connected outputs, later-feedback terms, visits, covariance traces, low owners, rational owners, and the R-063 forest have been assembled exactly once. The remaining question was whether stationary subtraction and a different global coordinate yield a genuinely smaller, refinement-safe theorem rather than another reformulation of the same obstruction.

Ten proof rounds were run with the earlier R-094–R-123 loop audit kept in view. The main correction is imposed at the start. R-121 supersedes the old R-119 Section 6 arithmetic-companion inference. In repository curl convention,

$$\text{curl } \omega_K = -\frac{40}{729}, \quad \text{curl } \omega_M = \frac{2720}{729}, \quad \text{curl } \omega_{M+K} = \frac{2680}{729}. \quad (1.2)$$

There is no structurally required $+40/729$ current. The surviving connection must be retained in the complete endpoint scalar.

R-124 conclusion. Stationary subtraction has an exact conditional symmetric-secant form. It telescopes pathwise under every representation-preserving same-root visit subdivision, provided each subvisit uses its moving endpoint and the reveal field, heat, covariance, and contraction-closed projection are unchanged. The formula exposes two favourable negative squares but does not give them a sign after their load-bearing mixed covariance is separated.

For a scalar fresh root, conditional replicas give an exact Hermite secant theorem. On the genuine R-123 first-linear-row cluster this closes the trace defect with the sharp unrestricted source coefficient $\kappa^2 = 6/(125P)$, so the action uses only $\eta_{\text{row}} = \kappa^2/2 = 3/(125P) < 9/20$, no sextic payment, and no feedback derivatives. Finite Hermite degree gives a strictly smaller sharp coefficient. This is a real positive theorem, but replica symmetrisation alone does not bound the full rational/Cartan/forest secant.

Four broad bypasses are now decisively excluded or parked. The surviving Cartan block cannot reuse the zeroth-order $H^{-11/10}$ coefficient class. Expected source and sextic budgets do not imply the separated $10/3$ or fifth moments demanded by coefficient-first Carleson/BMO estimates. A covariant connection square leaves a negative acceleration owner. Generic OU/Follmer rewritings expose or restate the trace debt rather than cancel it.

The next noncyclic theorem is therefore a coefficient-level, owner-complete, correlated signed root-shell estimate for the stationary secant itself, together with a separate cutoff-uniform sum bound for its owner-complete stationary baseline. The cutoff-uniform production inequality, $\text{OVERLAP}_{\text{src}}$, Nelson, removals, the interacting measure, and Sector-A closure remain open. Tier stays T4.

2. Round two: exact conditional stationary polarization

Let $\mathcal{G}$ be the strict-past sigma-field and let H be a finite or separable real or complex Hilbert space. Assume

$$\Phi_u \in L^2(\Omega; H), \quad \Theta_u \in L^1(\Omega), \quad u \in \{0, h\}.$$

Write $\mathbb{E}_0 = \mathbb{E}[\cdot \mid \mathcal{G}]$ and define

$$b_u = \mathbb{E}_0 \Phi_u, \quad Y_u = \Phi_u - b_u, \quad (2.1)$$

$$D_{0,u} = \mathbb{E}_0 \{\Theta_u - \|Y_u\|^2\}, \quad S_u = D_{0,u} - \|b_u\|^2. \quad (2.2)$$

Then

$$\boxed{S_u = \mathbb{E}_0 \{\Theta_u - \|\Phi_u\|^2\}.} \quad (2.3)$$

Put

$$\Delta\Phi = \Phi_h - \Phi_0, \quad \Delta\Theta = \Theta_h - \Theta_0. \quad (2.4)$$

Theorem 2.1 (stationary-polarized trace secant)

Almost surely,

$$\boxed{S_h - S_0 = \mathbb{E}_0 [\Delta\Theta - \text{Re}\langle\Phi_h + \Phi_0, \Delta\Phi\rangle].} \quad (2.5)$$

Equivalently,

$$\boxed{S_h - S_0 = \mathbb{E}_0 [\Delta\Theta - 2\text{Re}\langle\Phi_0, \Delta\Phi\rangle - \|\Delta\Phi\|^2].} \quad (2.6)$$

Proof. The real polarization identity gives

$$\|\Phi_h\|^2 - \|\Phi_0\|^2 = \text{Re}\langle\Phi_h + \Phi_0, \Phi_h - \Phi_0\rangle = 2\text{Re}\langle\Phi_0, \Delta\Phi\rangle + \|\Delta\Phi\|^2. \quad (2.7)$$

Subtract (2.3) at $u = 0$ from (2.3) at $u = h$ and use (2.7). No derivative of h is taken. $\square$

The symmetric quantity

$$\Psi_{h|0} = \Delta\Theta - \text{Re}\langle\Phi_h + \Phi_0, \Delta\Phi\rangle \quad (2.8)$$

is the smallest two-coordinate endpoint form. Formula (2.6) is preferable for estimates because its last term is a favourable negative square.

With $\Delta b = b_h - b_0$ and $\Delta Y = Y_h - Y_0$, the exact centered form is

$$\begin{aligned} S_h - S_0 &= \mathbb{E}_0 \Delta\Theta - 2\text{Re}\langle b_0, \Delta b\rangle - \|\Delta b\|^2 \\ &\quad - 2\text{Re}\mathbb{E}_0 \langle Y_0, \Delta Y\rangle - \mathbb{E}_0 \|\Delta Y\|^2. \end{aligned} \quad (2.9)$$

The mixed centered covariance in the second line is load-bearing. Replacing it by conditional means is illegal because Φ_0 is not generally $\mathcal{G}$ -measurable.

R-120 proves $S_0 = 0$ only for its uncontrolled raw six-row stationary scope. It does not prove rootwise cancellation of the owner-complete rational/Cartan/forest baseline. Thus S_0 remains in (2.5) until a global owner-complete stationary identity or uniform baseline bound is proved.

3. Round three: exact visit-subdivision quotient

Consider a legal same-root chain

$$(\Phi_0, \Theta_0), (\Phi_1, \Theta_1), \dots, (\Phi_L, \Theta_L), \quad (3.1)$$

and put

$$\Delta_\ell \Phi = \Phi_\ell - \Phi_{\ell-1}, \quad \Delta_\ell \Theta = \Theta_\ell - \Theta_{\ell-1}, \quad (3.2)$$

$$\Psi_\ell = \Delta_\ell \Theta - \text{Re}\langle\Phi_\ell + \Phi_{\ell-1}, \Delta_\ell \Phi\rangle. \quad (3.3)$$

Theorem 3.1 (moving-endpoint subdivision invariance)

Pathwise,

$$\boxed{\sum_{\ell=1}^L \Psi_\ell = \Theta_L - \Theta_0 - (\|\Phi_L\|^2 - \|\Phi_0\|^2).} \quad (3.4)$$

Proof. Both the trace differences and the squared-norm differences telescope in (3.3). $\square$

This is the R-116 endpoint quotient in the stationary-subtracted coordinate. For two increments f and ι , the moving-baseline form contains

$$-2 \operatorname{Re}\langle \Phi_0, f \rangle - \|f\|^2 - 2 \operatorname{Re}\langle \Phi_0 + f, \iota \rangle - \|\iota\|^2, \quad (3.5)$$

so the f - ι cross appears exactly once. Reusing the fixed Φ_0 in the second subvisit drops

$$-2 \operatorname{Re}\langle f, \iota \rangle. \quad (3.6)$$

The production use of Theorem 3.1 requires the same fresh root and reveal sigma-field, the same target heat and covariance, the same contraction-closed projection, and no newly available child control. A refinement that reveals a new independent coordinate is a new chart, not a representation-preserving subdivision. R-104's broader chart invariance is safe only after complete owner/root/visit aggregation and outer expectation.

Full covariance is also mandatory. For redundant coefficients $(1/2, 1/2)$ and covariance matrix with every entry equal to one, the true trace is one, whereas diagonal-only accounting gives $1/2$.

4. Round four: owner-complete incidence and the surviving connection

For the legal R-119 two-block/two-visit chart, let the fixed R-120 six-row endpoint and primitive trace be

$$\Phi_{H,\mathcal{C}} = \left(\Pi_{\mathcal{C}} \mathbb{E}_{\text{future}} C_6(\widehat{W_{H,r}}) V_{H,i}(n) \right)_{i,n \in \mathcal{C}}, \quad (4.1)$$

$$\Theta_{H,\mathcal{C}} = \mathbb{E}_{\text{future},G} \sum_{i,n \in \mathcal{C}} \int \left| \Pi_{\mathcal{C}} C_6(\widehat{W_{H,r}}) \partial_i G(n) \right|^2 \mu_{\Sigma}(\mathrm{d}r). \quad (4.2)$$

Every $\mathcal{C}$ is contraction closed. Controlled and stationary endpoints use the same heat/covariance coupling.

The endpoint pair in Theorem 2.1 represents production only after retaining: all six rows; all contraction-connected outputs; the primitive trace once or the R-063 forest expansion once, never both; every later-feedback cross; the full $R_Q + M_U + K_R$ rational endpoint; the conditional and complete low owners; the paid difference; and one REG Cartan FAR incidence. The outer expectation is essential when the strict-past baseline is random.

R-121 proves the rational endpoint identity

$$\sum_k (R_{Q,k} + M_{U,k} + K_{R,k}) = \frac{1}{2} B_L : Q_L - \frac{1}{2} B_0 : Q_0. \quad (4.3)$$

R-121 states the two-visit case. The displayed finite- L form follows by successively applying that identity to adjacent endpoint pairs; every intermediate $B_\ell : Q_\ell$ occurs once with each sign and telescopes. This is pathwise and subdivision invariant after recombination. Its covariance-normal trace and the R-123 primitive Θ remain distinct until a coefficient-level identity relates them. Appending the R-063 forest to both would double count the same owner.

The exact R-121 curl values (1.2) show that stationary polarization does not cancel the Cartan block. It places the surviving 2680/729 connection in the correct endpoint scalar. Any future reconstruction that inserts a +40/729 current fails this checksum.

5. Round five: critical resonance blocks zeroth-order reuse

R-120 controls a zeroth-order variable multiplier against an $H^{-11/10}$ coefficient. The surviving first-order Cartan form has a different sharp boundary. On a normalized one-dimensional torus embedded in $\mathbb{T}^3$, take

$$z_N = (1, N^{-2} \sin Nx), \quad Q_N = N^s \cos Nx, \quad (5.1)$$

and pair against the fixed skew generator. Then

$$\|z_N\|_{H^2}^2 = \frac{3}{2} + \frac{1}{2N^2} + \frac{1}{2N^4}, \quad \|Q_N\|_{H^{-s}}^2 = \frac{N^{2s}}{2(1+N^2)^s} \leq \frac{1}{2}, \quad (5.2)$$

while

$$\langle Q_N, z_N^T J \partial_x z_N \rangle = -\frac{1}{2} N^{s-1}. \quad (5.3)$$

The L^6 norm of z_N is uniformly bounded. At $s = 11/10$, (5.3) diverges as $N^{1/10}$. Thus the R-120 zeroth-order coefficient class cannot control the R-121 first-order connection.

R-063 does contain balanced stationary Gaussian-jet gains, but only for its declared stationary/deterministic-translation scope. It does not identify the same-root adapted total symbol. Critical-face vanishing, support-only separation, or a universal input-leaf cancellation is therefore not an available production theorem.

6. Round six: rare-event obstruction to separated Carleson moments

Let E be a predictable event of probability $p = N^{-6}$ and let the future bracket be

$$B_N = N^6 \mathbf{1}_E. \quad (6.1)$$

Then

$$\mathbb{E} B_N = 1, \quad \mathbb{E} B_N^{5/3} = N^4, \quad \mathbb{E} B_N^{5/2} = N^9. \quad (6.2)$$

The first divergence defeats deriving the R-120 10/3 rough input from expected source/sextic budgets; the second defeats the separated fifth moment needed by the R-121 Cartan estimate. This is the same failure class as the registered weighted-Doob and translated-model rare-event no-gos, now applied to the R-123 trace-excess successor.

Equation (6.2) is a method obstruction, not a counterexample to the complete production packet. It says that expectation must remain inside the exact correlated signed root-shell object. An essential-supremum BMO norm or a coefficient-first positive moment cannot be inferred from the available expected budgets.

7. Round seven: the signed aggregate route and its acceptance test

For a full chart π , define the stationary secant only after complete within-chart aggregation:

$$\mathfrak{S}_{J,\pi}(h) = \sum_{k,\mathcal{C}} \mathbb{E} [\Delta \Theta_{k,\mathcal{C}} - \text{Re} \langle \Phi_{h,k,\mathcal{C}} + \Phi_{0,k,\mathcal{C}}, \Delta \Phi_{k,\mathcal{C}} \rangle]. \quad (7.1)$$

By Theorems 2.1 and 3.1 this is exactly the difference of the R-123 trace excesses and has no visit-count or representation-depth factor.

The next sufficient estimate is

$$\boxed{\mathfrak{S}_{J,\pi}(h) \leq 2\eta \mathbb{E} \sum_k \|h_k\|^2 + 2\zeta \mathbb{E} \|Z_K\|_6^6 + 2C_{\text{corr}},} \quad (7.2)$$

with $\eta < 9/20$, $\zeta < 3/20$, and constants uniform in cutoff, chart, control, dimension, block count, visit count, and representation-preserving refinement. The global owner-complete stationary baseline must separately satisfy

$$\sum_{k,\mathcal{C}} \mathbb{E} S_{0,k,\mathcal{C}} \leq 2C_0. \quad (7.3)$$

Then (7.2) yields the original R-123 controlled trace-excess target with additive constant $2(C_{\text{corr}} + C_0)$. R-093's Gaussian centering may be invoked only after the full owner sum, not rootwise from R-120.

Calling (7.2) a “signed Carleson estimate” does not prove it. A valid proof must first construct the coefficient-level identity that:

1. matches the R-121 $B : Q$ endpoint with the R-123 primitive-trace allocation;
2. expands the adapted R-063 forest exactly once;
3. retains the 2680/729 connection and the R-120 zeroth-order multipliers in one quotient-basic total symbol;
4. produces a spatial/root gain inside the correlated expectation, before any positive part or separate rough moment.

Without these four items, a conventional Carleson theorem either invokes false separated moments from Section 6 or merely renames the open inequality (7.2).

8. Round eight: replica/Hermite normal form

Condition on the strict past and take one scalar fresh root. For a finite Hermite endpoint law write

$$\Phi_h = \sum_{n=0}^M f_n^h H_n(\xi), \quad \theta_h = \mathbb{E}_0 \Theta_h, \quad (8.1)$$

with probabilists' unnormalised Hermites, $\mathbb{E} H_n^2 = n!$. Let Φ'_h be a conditionally independent replica. Then

$$\frac{1}{2} \mathbb{E}_0 \|\Phi_h - \Phi'_h\|^2 = \sum_{n=1}^M n! \|f_n^h\|^2, \quad (8.2)$$

and

$$S_h = \theta_h - \sum_{n=0}^M n! \|f_n^h\|^2. \quad (8.3)$$

For $g_n = f_n^h - f_n^0$,

$$S_h - S_0 = \theta_h - \theta_0 - 2 \sum_{n=0}^M n! \operatorname{Re} \langle f_n^0, g_n \rangle - \sum_{n=0}^M n! \|g_n\|^2. \quad (8.4)$$

Thus replicas supply precisely the $n \geq 1$ centred-variance part of the last favourable square. The $n = 0$ term is instead the conditional-mean reserve from stationary polarization. Neither part controls the signed trace/current cross in the first line. For the full six-row cluster, replica symmetrisation alone is an exact restatement of (7.1).

9. Round nine: sharp legal first-linear-row theorem

For the R-123 normalized legal first-linear row,

$$\Phi_h = \kappa \left\{ \frac{\xi^2}{\sqrt{2}} e_{s2} + \xi h(\xi) e_c \right\}, \quad \Theta_h = \kappa^2 \left(\frac{\xi^2}{2} + h^2 + 1 \right), \quad (9.1)$$

where

$$P = 4 + 10^{-12}, \quad \kappa^2 = \frac{6}{125P}. \quad (9.2)$$

The stationary and control output coordinates are orthogonal. Hence

$$\boxed{S_h - S_0 = \kappa^2 \mathbb{E}[(1 - \xi^2)h^2] \leq \kappa^2 \mathbb{E}h^2.} \quad (9.3)$$

The corresponding action allocation is

$$\boxed{\eta_{\text{row}} = \frac{\kappa^2}{2} = \frac{3}{125P} \simeq 0.006 < \frac{9}{20}.} \quad (9.4)$$

No sextic payment, Dh , or D^2h is used.

There is also a sharp finite-degree theorem. If $h = \sum_{n=0}^M a_n H_n$, then

$$\mathbb{E}[(\xi^2 - 1)h^2] = 2 \sum_{n=0}^M n n! a_n^2 + 2 \sum_{n=0}^{M-2} (n+2)! a_n a_{n+2}. \quad (9.5)$$

Let μ_M be the smallest eigenvalue of the parity-tridiagonal matrix

$$A_{nn} = 2n + 1, \quad A_{n,n+2} = \sqrt{(n+1)(n+2)}. \quad (9.6)$$

Indeed, for the normalized basis $e_n = H_n / \sqrt{n!}$,

$$\xi e_n = \sqrt{n+1} e_{n+1} + \sqrt{n} e_{n-1},$$

and hence

$$\xi^2 e_n = \sqrt{(n+1)(n+2)} e_{n+2} + (2n+1) e_n + \sqrt{n(n-1)} e_{n-2}. \quad (9.6a)$$

Thus A represents multiplication by ξ^2 on the degree- M Hermite space, while $A - I$ represents the quadratic form in (9.5). Rayleigh–Ritz gives

$$\boxed{S_h - S_0 \leq \kappa^2 (1 - \mu_M) \mathbb{E}h^2.} \quad (9.7)$$

Under $y = \xi^2/2$, its even and odd blocks are the Jacobi matrices for $2y$ with Laguerre weights $y^{-1/2}e^{-y}$ and $y^{1/2}e^{-y}$. Consequently their characteristic polynomials are the corresponding monic multiples of $L_d^{-1/2}(\lambda/2)$ and $L_d^{1/2}(\lambda/2)$. For odd M , monotonicity of Laguerre zeros in the parameter puts the even minimum first. For even M , the derivative identity $(L_d^{-1/2})' = -L_{d-1}^{1/2}$ and strict zero interlacing give the same ordering. Therefore

$$\mu_M = 2z_{\min} \left(L_{\lfloor M/2 \rfloor + 1}^{-1/2} \right) > 0. \quad (9.8)$$

In particular, $M = 0, 1$ gives coefficient zero, while

$$\mu_2 = \mu_3 = 3 - \sqrt{6}, \quad 1 - \mu_2 = 1 - \mu_3 = \sqrt{6} - 2. \quad (9.9)$$

The unrestricted coefficient κ^2 is sharp. Smooth Gaussian bumps $h_a(\xi) = e^{-a\xi^2}$ satisfy

$$\frac{\mathbb{E}[(1 - \xi^2)h_a^2]}{\mathbb{E}h_a^2} = \frac{4a}{1 + 4a} \rightarrow 1. \quad (9.10)$$

Scaling the bump amplitude to zero makes its homogeneous sextic/source ratio tend to zero, so no homogeneous sextic correction uniformly improves the source coefficient below κ^2 .

The old derivative-secant obstruction disappears on this row. For

$$h_N(\xi) = \frac{\sin(N\xi)}{N}, \quad (9.11)$$

one has

$$\mathbb{E}h_N^2 = \frac{1 - e^{-2N^2}}{2N^2}, \quad (9.12)$$

$$\mathbb{E}|h_N|^6 = \frac{10 - 15e^{-2N^2} + 6e^{-8N^2} - e^{-18N^2}}{32N^6}, \quad (9.13)$$

$$S_{h_N} - S_0 = -2\kappa^2 e^{-2N^2}, \quad D_0(h_N) - D_0(0) = \kappa^2(e^{-N^2} - 2e^{-2N^2}). \quad (9.14)$$

Replica/secant control sees the bounded law, not the growing derivative.

Automatic nonpositivity is nevertheless false. For the bounded smooth law $h = d \cos(t\xi)$,

$$\boxed{S_h - S_0 = 2\kappa^2 d^2 t^2 e^{-2t^2} > 0.} \quad (9.15)$$

This is the exact evidence for NG-2026-07-30-A13-REPLICA-VARIANCE-AUTOMATIC-TRACE-DOMINATION. It refutes only automatic replica domination, not the source-paid theorem (9.3) or the full A1 objective.

10. Round ten: covariant-Hessian and OU/Follmer fallbacks

10.1 Covariant connection-square audit

For the scalar current ray write

$$E(U) = \frac{1}{2} \int g(U) U_x^2, \quad \Gamma(U) = \frac{g'(U)}{2g(U)}, \quad \nabla_x z = z_x + \Gamma(U) U_x z. \quad (10.1)$$

Direct differentiation gives the exact identity

$$\boxed{D^2 E(U)[z, z] = \int g(U) |\nabla_x z|^2 + DE(U)[\Gamma(U) z^2].} \quad (10.2)$$

The second term is the connection acceleration owner and has no fixed sign. For the homogeneous leading coefficient $g(U) = 4CU^2$, with normalized Haar measure,

$$U = H(2 + \cos x), \quad z = 2 - \cos x, \quad (10.3)$$

one obtains exactly

$$\int g |\nabla_x z|^2 = 2CH^2, \quad DE[\Gamma z^2] = -15CH^2, \quad D^2 E = -13CH^2. \quad (10.4)$$

For the A1 large-amplitude coefficient

$$C = c_0 + c_1(4/9)^2 = \frac{9}{500P}, \quad (10.5)$$

this is the R-123 leading term $-117H^2/(500P)$, with bounded positive-floor corrections. The ray is one-dimensional and has zero Riemann curvature. A connection square alone therefore cannot restore positivity; the negative term moves into acceleration. A pure square can apply at a critical base $DE = 0$ or along a target-geodesic variation, neither of which is available for arbitrary adapted production controls.

10.2 Exact OU Hessian representation

Let the higher-chaos residual be

$$R = \sum_{n \geq 2} I_n(r_n), \quad P_s I_n = e^{-ns} I_n. \quad (10.6)$$

Then

$$\mathbb{E} \|D^2 I_n(r_n)\|_{\text{HS}}^2 = n(n-1)n! \|r_n\|^2. \quad (10.7)$$

Consequently the R-119 bare zero-chaos debt has the exact form

$$\Pi_0 \mathcal{R}_{\text{bare}} = -\frac{1}{2} \sum_{n \geq 2} (n-1)n! \|r_n\|^2 = -\int_0^\infty \mathbb{E} \|D^2 P_s R\|_{\text{HS}}^2 ds. \quad (10.8)$$

For $R = H_2/2 - H_3/3 + H_4/4$, both sides equal $-19/6$. OU integration exposes a nonpositive curvature energy, strictly negative when $R \neq 0$; it does not cancel it. Low/output/R-063 companions remain mandatory.

The total Gibbs thermodynamic/Follmer identity is still the unknown endpoint difference, as registered by the total-time-integration no-go. A generic all-law pointwise estimate would again force the nonintegrable coefficient $b(t) \geq 3/t$. Only a genuinely Gibbs-specific, root-local, time-integrated signed estimate remains as a fallback after the direct correlated secant route, not as a replacement for it.

11. Ten-round synthesis: strongest theorem and noncyclic roadmap

The ten-round evidence map is

round	verdict	reusable conclusion
1	corrected	+40/729 search retired; 2680/729 retained
2	advanced	exact conditional stationary secant
3	advanced	moving-endpoint subdivision quotient
4	advanced	owner firewall and trace distinction
5	failed shortcut	$H^{-11/10}$ cannot pay first-order Cartan
6	failed shortcut	separated rough moments fail on rare events
7	scoped	four acceptance tests for a signed root-shell theorem
8	mixed	replicas give a square but leave the signed secant
9	advanced	sharp legal-row source theorem
10	mixed	covariant/OU identities expose, not remove, the debt.

(11.1)

The proof programme has therefore advanced in three concrete ways. It has a minimal refinement-invariant target, an exactly closed nontrivial adapted row with a sharp source constant, and decisive stop tests excluding the separated-moment implementations that would otherwise repeat the prior cycle. The next work item is not another local coordinate or positive normalizer. It is the coefficient-level reconstruction listed after (7.2), followed by one correlated signed root-shell estimate with all sums inside expectation and the separate uniform stationary-baseline bound (7.3).

If that reconstruction fails, the failure must identify a concrete mismatch between the R-121 rational endpoint and the R-123 primitive trace, or a non-summable surviving total symbol. Only then should the Gibbs-specific time-integrated fallback be resumed. Replica, covariant connection, generic Follmer, BMO, and coefficient-first Carleson routes are not to be reopened without a new hypothesis that defeats their displayed stop tests.

12. Executable verification

The primary SymPy executable derives the A1 constants from the production manifest and checks stationary polarization, complex Hilbert polarization, conditional cells, centered covariance, replicas, subdivision, full covariance, Hermite forms and sharp constants, the legal-row fixtures, all three R-121 curls, critical regularity, rare events, covariant Hessian, and the OU identity. It passes 55/55 assertions.

The non-importing independent executable uses standard-library exact Fractions, monomial Gaussian moments, a bivariate rational jet, Simpson Gaussian quadrature, direct rare-event enumeration, and alias-free torus quadrature. It independently passes 55/55 assertions. The integrated verifier reruns both children, checks deterministic stored artefacts, pins all authorities, checks the note, PDF, manifest, exploration records, negative authority, and public surfaces, and rejects a stale package.

13. Devil’s-advocate review

1. **Objection: stationary subtraction proves the full baseline is zero. UPHELD AGAINST THE CLAIM.** R-120 proves that only for its raw uncontrolled six-row scope. The owner-complete baseline stays until global aggregation.
2. **Objection: visit invariance permits a fixed stationary baseline at every subvisit. UPHELD AGAINST THE CLAIM.** Equation (3.6) is the missing cross. Every subvisit must use its moving endpoint.
3. **Objection: replicas make the trace defect nonpositive. UPHELD AGAINST THE CLAIM.** Equation (9.15) is strictly positive. Replicas supply a negative square, not domination of the trace secant.
4. **Objection: the legal-row coefficient can be made uniformly smaller by a sextic term. UPHELD AGAINST THE CLAIM.** Gaussian bumps approach the sharp coefficient and amplitude scaling removes the homogeneous sextic correction.
5. **Objection: the R-120 $H^{-11/10}$ moment pays Cartan as well. UPHELD AGAINST THE CLAIM.** Equation (5.3) diverges at that regularity.
6. **Objection: expected budgets imply the required BMO or rough moments. UPHELD AGAINST THE CLAIM.** Equation (6.2) keeps the mean fixed while both separated moments diverge.
7. **Objection: a Levi–Civita connection repairs raw-Hessian positivity. UPHELD AGAINST THE CLAIM.** The acceleration owner in (10.4) is $-15CH^2$ and dominates the $2CH^2$ connection square.
8. **Objection: the OU identity is the missing cancellation. UPHELD AGAINST THE CLAIM.** Equation (10.8) is nonpositive and exactly equals the existing bare debt. It is a diagnostic representation.

14. Result footer

Result ID: A13-CLASSII-STATIONARY-POLARIZED-TRACE-DEFECT-
REPLICA-ROOT-SHELL-BOUNDARY (R-124).

Precise statement: Theorem 2.1, Theorem 3.1, Eqs. (8.2)–(8.4),
sharp legal-row Eqs. (9.3)–(9.10), and the exact method boundaries
in Sections 4–7 and 10.

Scope: fixed finite cutoff and positive rational floor for the A13

production interpretation; abstract Hilbert conditional identities;
 scalar finite-Hermite and legal-row sharp theorem; homogeneous
 covariant-Hessian fixture; no cutoff-uniform production estimate.
 Dependencies: A1; R-063, R-093, R-104, R-108, R-116,
 R-118, R-119, R-120, R-121, R-122, R-123.
 Evidence grade: ANALYTIC, EXACT, EXECUTED.
 Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe
 codes/foundations/
 a13_classii_stationary_polarized_trace_defect_replica_root_shell_
 boundary_verify.py
 Expected output: R-124 integrated PASS; primary 55/55 and independent
 55/55 children PASS; all authority, source, note, PDF, manifest,
 exploration, negative-result, and public-surface checks pass.
 Falsification gate: any failed polarization, conditional, subdivision,
 covariance, Hermite, legal-row, curl, critical, rare-event, covariant,
 OU, child/source/note/PDF/manifest, or scope assertion.
 Tier before / after: T4 / T4.
 No-overclaim statement: R-124 does not prove the owner-complete
 cutoff-uniform correlated root-shell estimate, OVERLAP_src, Nelson,
 removals, the interacting measure, Sector-A closure, or T5-T7.
 Next required action: construct the coefficient-level identity linking
 the R-121 rational endpoint, R-123 primitive trace, and adapted R-063
 forest exactly once, then prove Eqs. (7.2)--(7.3) with root and shell
 sums retained inside the signed expectation.